

OLEN'YA RECHKA, Russian Federation

WMO: 299740

Lat: 52.8051N

Lon: 93.2406E

Elev: 1399

StdP: 85.61

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.9	-25.6	-30.7	0.3	-27.4	-28.7	0.3	-25.0	14.0	-18.1	12.3	-18.8	2.8	90	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	22.8	14.2	21.1	13.6	19.4	12.9	15.6	20.3	14.7	19.5	13.8	18.2	2.6	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.9	11.7	17.2	12.8	10.9	16.3	11.9	10.3	15.5	48.9	20.4	46.0	19.7	43.3	18.3	19.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.0	8.8	7.8	-31.9	26.8	4.5	1.5	-35.1	27.9	-37.7	28.8	-40.2	29.6	-43.5	30.7		
			-32.0	17.6	4.5	1.0	-35.2	18.3	-37.8	18.9	-40.3	19.4	-43.5	20.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.1	-17.5	-15.0	-7.7	-1.6	3.6	10.7	13.0	11.6	5.9	-2.3	-11.6	-14.8	(6)
(7)		DBStd	11.69	4.11	5.12	5.48	5.23	5.09	4.66	3.14	3.67	5.10	5.40	5.87	4.38	(7)
(8)		HDD10.0	4679	851	701	549	348	208	48	8	25	144	381	649	768	(8)
(9)		HDD18.3	7451	1109	934	807	597	456	230	168	211	374	638	899	1026	(9)
(10)		CDD10.0	276	0	0	0	0	10	69	101	74	20	1	0	0	(10)
(11)		CDD18.3	6	0	0	0	0	0	2	2	2	0	0	0	0	(11)
(12)		CDH23.3	35	0	0	0	0	1	14	13	7	0	0	0	0	(12)
(13)		CDH26.7	1	0	0	0	0	0	1	1	0	0	0	0	0	(13)
(14)	Wind	WSAvg	2.5	4.5	3.4	2.2	1.9	1.8	1.7	1.5	1.6	1.8	2.1	3.3	4.4	(14)
(15)	Precipitation	PrecAvg	1228	62	39	54	103	116	128	172	159	120	100	93	72	(15)
(16)		PrecMax	1756	567	90	98	196	221	284	356	300	212	216	204	223	(16)
(17)		PrecMin	903	8	8	18	35	50	20	19	26	47	38	30	20	(17)
(18)		PrecStd	175	92	20	21	47	44	52	66	56	38	44	41	42	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.5	-1.3	6.7	12.7	20.1	25.1	25.1	24.3	20.9	12.6	3.1	-4.0	(19)
(20)			MCWB	-8.5	-2.5	1.5	4.8	9.6	14.9	16.3	14.7	11.4	5.2	0.1	-4.8	(20)
(21)		2%	DB	-9.1	-4.3	4.2	10.1	16.7	22.6	22.6	21.7	18.0	9.7	0.0	-6.6	(21)
(22)			MCWB	-10.0	-5.7	-0.1	3.4	8.6	13.5	15.0	14.2	10.5	4.0	-2.3	-7.6	(22)
(23)		5%	DB	-10.7	-6.5	2.0	8.2	14.0	20.8	20.7	19.8	15.6	7.5	-2.1	-7.8	(23)
(24)			MCWB	-11.4	-8.0	-1.6	2.4	7.1	13.1	14.3	13.1	9.7	3.1	-4.2	-8.6	(24)
(25)		10%	DB	-12.4	-8.3	-0.2	6.1	11.6	18.7	18.9	17.9	13.2	5.2	-4.3	-9.1	(25)
(26)			MCWB	-13.1	-9.6	-2.9	1.3	5.8	12.4	13.6	12.5	8.4	2.0	-5.5	-9.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.4	-2.2	1.6	5.9	10.9	16.6	17.3	16.0	12.6	6.5	0.1	-4.8	(27)
(28)			MCDWB	-7.9	-1.7	5.7	11.3	18.1	21.6	22.6	20.8	18.4	9.8	2.4	-4.3	(28)
(29)		2%	WB	-9.9	-5.5	0.2	4.1	9.1	15.0	15.9	14.9	11.3	4.8	-2.2	-7.4	(29)
(30)			MCDWB	-9.2	-4.6	3.5	8.9	15.5	20.5	20.5	20.1	16.5	8.3	-0.4	-6.8	(30)
(31)		5%	WB	-11.4	-7.9	-1.3	2.9	7.7	13.9	15.0	14.0	10.1	3.6	-4.0	-8.6	(31)
(32)			MCDWB	-10.7	-6.7	1.8	7.0	12.9	19.2	19.4	18.8	14.9	7.1	-2.4	-7.8	(32)
(33)		10%	WB	-13.0	-9.6	-2.9	1.8	6.3	13.0	14.1	13.0	8.8	2.2	-5.5	-9.8	(33)
(34)			MCDWB	-12.5	-8.3	-0.5	5.5	10.9	17.6	18.2	16.8	12.5	5.3	-4.2	-9.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.0	5.5	7.8	8.4	8.4	9.0	8.2	8.4	8.0	6.4	4.8	3.7	(35)
(36)			MCDBR	4.5	6.4	8.8	10.4	11.4	11.9	11.3	11.6	11.2	9.1	5.8	5.4	(36)
(37)			MCWBR	4.3	5.5	5.9	5.7	6.0	5.9	5.7	5.7	5.9	5.4	4.7	5.0	(37)
(38)		5% WB	MCDBR	4.5	6.5	8.4	9.2	10.5	10.4	9.4	10.3	10.2	8.2	5.2	5.2	(38)
(39)			MCWBR	4.3	5.6	5.9	5.4	6.0	5.7	5.3	5.4	5.7	5.2	4.4	4.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.253	0.285	0.327	0.366	0.374	0.362	0.365	0.352	0.323	0.312	0.285	0.256	(40)	
(41)		taud	2.098	2.067	2.030	2.131	2.241	2.388	2.403	2.417	2.450	2.330	2.122	2.037	(41)	
(42)		Ebn at Noon	722	807	844	861	874	889	879	872	860	780	663	624	(42)	
(43)		Edh at Noon	71	102	132	136	129	113	109	102	87	79	70	63	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.15	2.20	3.63	4.64	5.31	5.98	5.42	4.60	3.34	1.96	1.03	0.76	(44)	
(45)		RadStd	0.09	0.14	0.20	0.25	0.42	0.56	0.38	0.37	0.35	0.18	0.12	0.07	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page